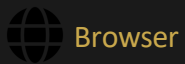


Agent · 3-Agent AI System · Browser + Python
CLI

SOW Orchestrator

Autonomous 3-agent pipeline · Discovery Analysis
Scope Validation · Professional SOW Drafting

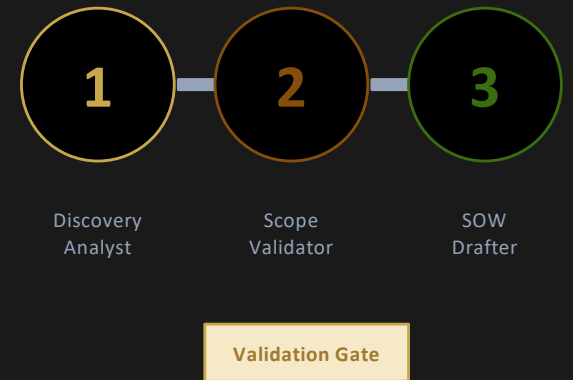


Browser



Python CLI

Daniel Mazzini · Principal Architect & Senior TPM



THE PROBLEM

Writing a SOW is one of the most time-consuming consulting deliverables



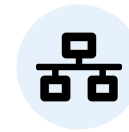
4-8 hours per SOW

A senior architect spends half a day converting discovery notes into a scope document — every single engagement, from scratch.



Scope errors reach customers

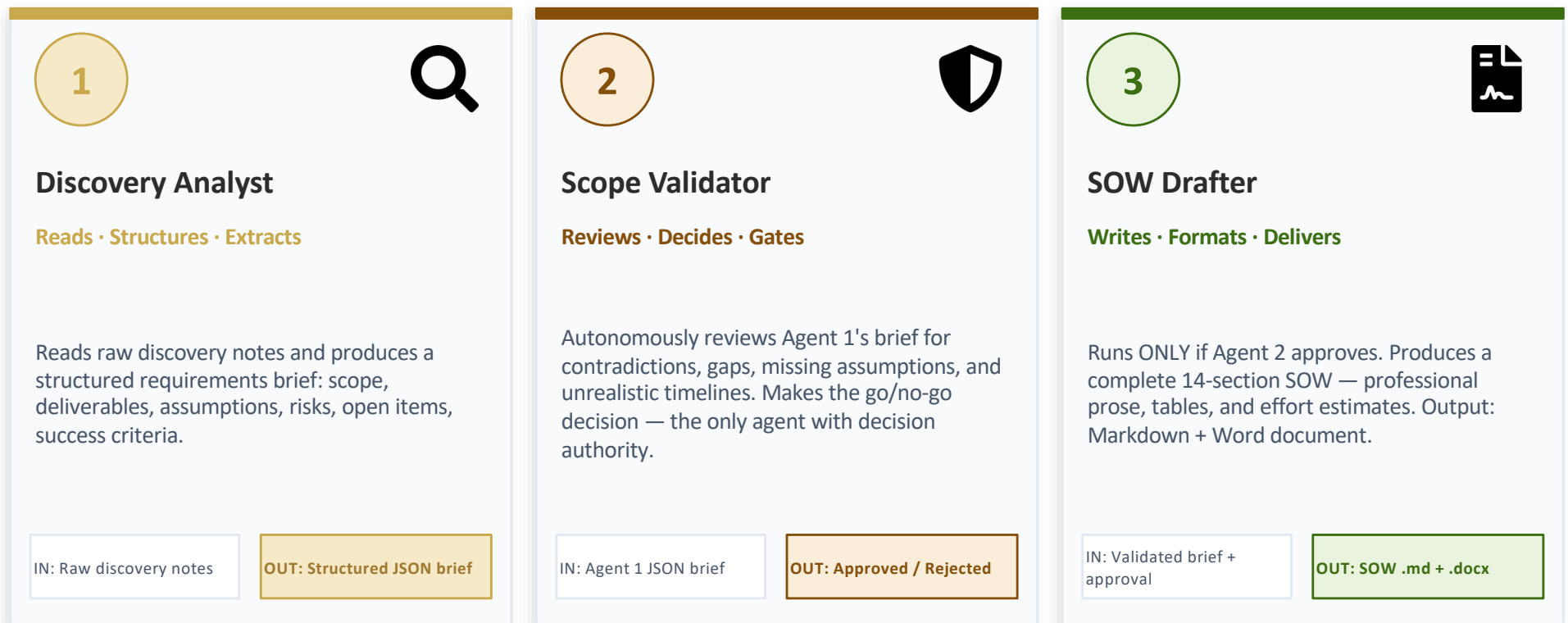
Contradictions, missing assumptions, and ambiguous deliverables are caught too late — after the SOW is already in the customer's hands.



No pipeline integration

SOW creation is a manual, isolated task. There's no way to plug it into a CI/CD pipeline, a CRM trigger, or a batch process.

Three specialized agents — one autonomous decision gate



HOW IT WORKS

From raw notes to signed SOW in under 60 seconds

1

Paste notes

Raw discovery notes — bullet points, stream of consciousness, anything. Browser textarea or text file for Python CLI.

2

Agent 1 runs

Extracts scope, deliverables, assumptions, risks, success criteria into structured JSON.

3

Agent 2 decides

Reviews brief autonomously. Returns: Approved, Approved with Warnings, or Rejected.

If rejected

If approved

System architecture — HLD



Single-file browser deployment OR pip install + python sow_generator.py · No server required

TWO INTERFACES

Same 3-agent pipeline. Choose your interface.

Browser · index.html

Audience	Anyone — no technical setup required
Install	None — open file in Chrome / Safari
Input	Paste notes into textarea
Output	HTML SOW rendered in browser
Export	Print to PDF · Copy to clipboard
Scenarios	6 pre-loaded · dropdown select
Best for	Live demos · customer workshops · interviews
API key	Entered once · stored in browser localStorage danvzla.github.io/ai-sow-orchestrator

Python CLI · sow_generator.py

Audience	Engineers · technical teams · CI/CD pipelines
Install	pip install -r requirements.txt
Input	Text file or stdin · argparse CLI
Output	Markdown .md + Word .docx files
Export	Formatted Word doc — customer-ready
Scenarios	6 built-in · or provide any text file
Best for	Automation · batch processing · integration

```
$ python sow_generator.py notes.txt --type "Telco NFV"
```

Agent design & data flow — LLD

AGENT 1 — DISCOVERY ANALYST	AGENT 2 — SCOPE VALIDATOR	AGENT 3 — SOW DRAFTER
Role injection: Senior Solutions Architect 20+ yrs	Role injection: Senior QA Architect & Risk Manager	Role injection: Senior Principal Consultant 20+ yrs
Input: Sanitized discovery notes	Input: Agent 1 JSON brief (stringified)	Input: Validated brief + warnings
Output: JSON brief — 13 structured fields	Checks: Contradictions · Gaps · Unrealistic	Output: 14-section SOW (Markdown + .docx)
Token limit: 4000 · concise structured output	Output: decision + issues + quality_score	Token limit: 8000 · full document generation
Recovery: 4-attempt JSON repair chain	Gate logic: rejected → stop · else → proceed	Trigger: Only runs if Agent 2 approves

⚡ Autonomous validation gate

Agent 2 makes autonomous go / no-go decisions

87

Quality score: 87/100 · Decision: APPROVED · 3 scope items confirmed · 1 warning flagged

WARNING · Dependency Risk

Ericsson equipment coexistence during 3-year phase-out creates integration complexity not captured in deliverables.

REJECTED · Quality score: 42/100 · 2 critical issues found · Agent 3 blocked

Contradiction

Zero downtime requirement conflicts with 180 VM migration scope

Fix: Add migration approach and maintenance window schedule to discovery notes.

Missing

No success criteria defined for NSX microsegmentation

Fix: Define criteria: % of workloads segmented, audit evidence format, acceptance sign-off.

KEY BENEFITS

Why this matters

45s

Raw notes to SOW draft

vs 4-8 hours manually

3

Specialized AI agents

each with one responsibility

2

Deployment interfaces

Browser · Python CLI

3

Output formats

HTML · Markdown · Word (.docx)



Autonomous quality gate

Agent 2 catches contradictions and scope gaps before Agent 3 writes a word. Prevents change orders and customer disputes.



Python CLI + Word export

Engineers can integrate SOW generation into any pipeline. Input: text file. Output: formatted Word document ready to send to a customer.



Built from real experience

6 scenarios map to VCF, NSX, Telco NFV, SASE, Cloud, and NOC programs — the actual engagements from 20+ years across VMware, Dell EMC, and Juniper.

DEMO SCENARIOS

6 pre-loaded scenarios — browser dropdown or Python CLI

01

VCF Private Cloud

Financial Services · \$500K–\$2M · 9 months

180 VM migration · vSAN stretched cluster · NSX from day 1

For: Cisco · Nutanix · Broadcom

02

NSX Security Hardening

Healthcare · \$150K–\$500K · 14 weeks

HIPAA deadline · DFW across 4 hospitals · Epic vendor approvals

For: Palo Alto · VMware · Healthcare

03

Telco NFV Transformation

Tier 2 Carrier · \$2M+ · 18 months

5G readiness · vEPC + vIMS + vFirewall · OSS/BSS integration

For: Nokia · Ericsson · Mavenir

04

SASE / SD-WAN Migration

Manufacturing · \$150K–\$500K · 6 months

6 plants incl. Mexico · SAP cutover deadline · Zero trust

For: Cisco · Palo Alto · VMware

05

Cloud Modernization

SaaS Technology · \$95K · 8 weeks

AWS \$840K/month · 180 microservices · FinOps framework

For: AWS · Google Cloud · NetApp

06

NOC Automation Program

Cable MSO · \$1.4M · 18 months

12 NOCs · MTTR 4.2hr target 90min · ServiceNow

For: Nokia · Ericsson · Kyndryl

```
$ python sow_generator.py notes.txt --type "Telco NFV Transformation" --industry "Telco / CSP"
```

LIVE DEMO AVAILABLE

Try it — two ways



danvzla.github.io/ai-sow-orchestrator



```
pip install -r requirements.txt && python sow_generator.py notes.txt
```

```
github.com/danvzla/ai-sow-orchestrator
```

Daniel Mazzini

Principal Architect & Senior TPM · Cloud & Infrastructure · Telco · Automation

danvzla@gmail.com · +1 678-428-3789 · Coppell, TX · Bilingual: English & Spanish

